

Stress Analysis

Report_Traffic_Light_without_Motor_ABS



Analyzed File:	TLA_without_motor.ipt
Autodesk Inventor Version:	2027 (Build 310192000, 192)
Creation Date:	2026-08-14, 오후 1:05
Study Author:	Kwon_Doyeong
Summary:	

Static Analysis:1

General objective and settings:

Design Objective	Single Point
Study Type	Static Analysis
Last Modification Date	2026-08-14, 오후 1:00
Model State	[Primary]
Detect and Eliminate Rigid Body Modes	No

iProperties

Summary

Title	
Subject	
Author	Kwon

Project

Part Number	TLA_without_motor
Designer	Kwon

Physical

Material	Generic
Density	1 g/cm ³
Mass	0.115262 kg
Area	44260.2 mm ²
Volume	115262 mm ³
Center of Gravity	x=-17.0088 mm y=30.4949 mm z=5.28203 mm

Note: Physical values could be different from Physical values used by FEA reported below.

Mesh settings:

Avg. Element Size (fraction of model diameter)	0.1
Min. Element Size (fraction of avg. size)	0.2
Grading Factor	1.5
Max. Turn Angle	60 deg
Create Curved Mesh Elements	Yes

Material(s)

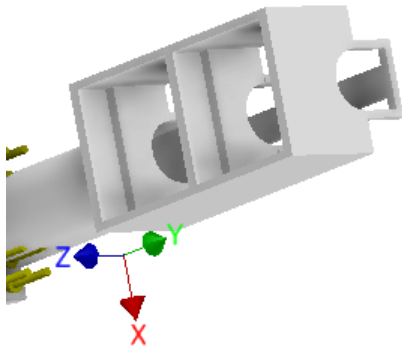
Name	ABS Plastic	
General	Mass Density	1.06 g/cm ³
	Yield Strength	20 MPa
	Ultimate Tensile Strength	29.6 MPa
Stress	Young's Modulus	2.24 GPa
	Poisson's Ratio	0.38 ul
	Shear Modulus	0.811594 GPa
Part Name(s)	TLA_without_motor.ipt	

Operating conditions

Gravity

Load Type	Gravity
Magnitude	9810.000 mm/s ²
Vector X	-0.000 mm/s ²
Vector Y	-9810.000 mm/s ²
Vector Z	-0.000 mm/s ²

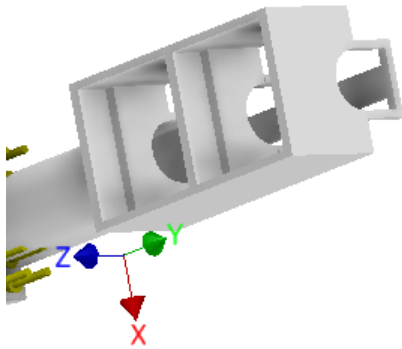
Selected Face(s)



Bearing Load:1

Load Type	Bearing Load
Magnitude	8.000 N
Vector X	0.000 N
Vector Y	-8.000 N
Vector Z	0.000 N

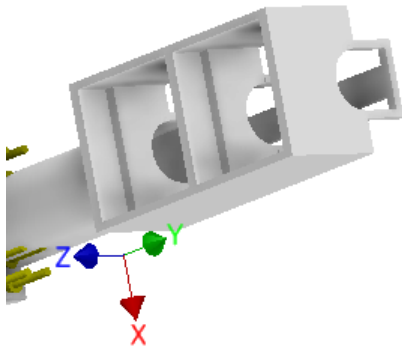
Selected Face(s)



Bearing Load:2

Load Type	Bearing Load
Magnitude	8.000 N
Vector X	0.000 N
Vector Y	-8.000 N
Vector Z	0.000 N

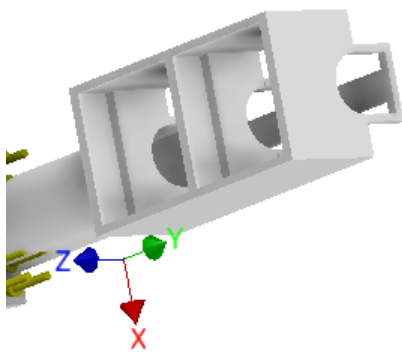
Selected Face(s)



Pressure:1

Load Type	Pressure
Magnitude	10.000 MPa

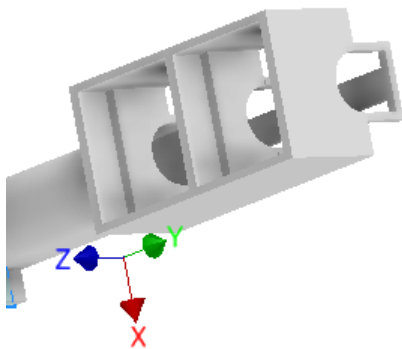
Selected Face(s)



Fixed Constraint:1

Constraint Type	Fixed Constraint
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Selected Face(s)



Results

Reaction Force and Moment on Constraints

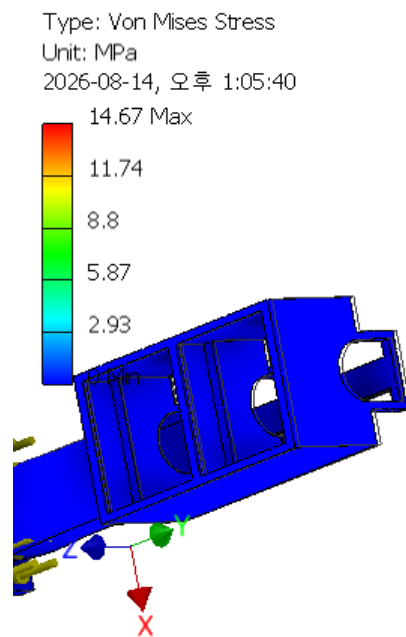
Constraint Name	Reaction Force		Reaction Moment	
	Magnitude	Component (X,Y,Z)	Magnitude	Component (X,Y,Z)
Fixed Constraint:1	742.761 N	0 N	2.27841 N m	2.27841 N m
		742.761 N		0 N m
		0 N		0 N m

Result Summary

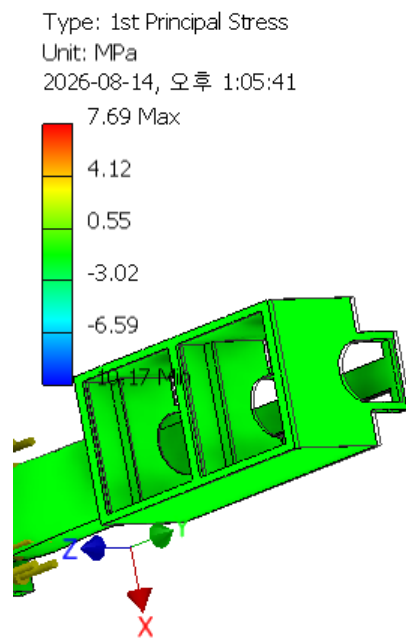
Name	Minimum	Maximum
Volume	31208.1 mm ³	
Mass	0.0330806 kg	
Von Mises Stress	0.0000289497 MPa	14.6725 MPa
1st Principal Stress	-10.1656 MPa	7.69009 MPa
3rd Principal Stress	-25.6316 MPa	2.94668 MPa
Displacement	0 mm	0.0258059 mm
Safety Factor	1.3631 ul	15 ul

Figures

Von Mises Stress

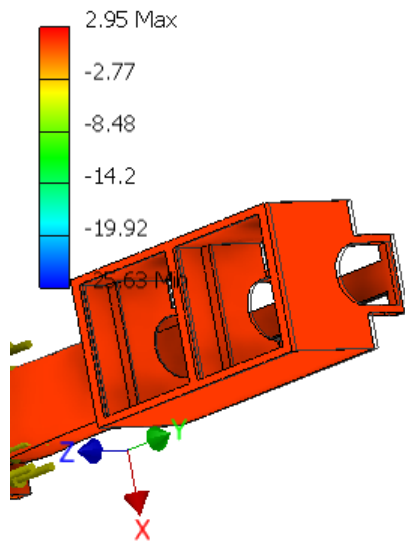


1st Principal Stress



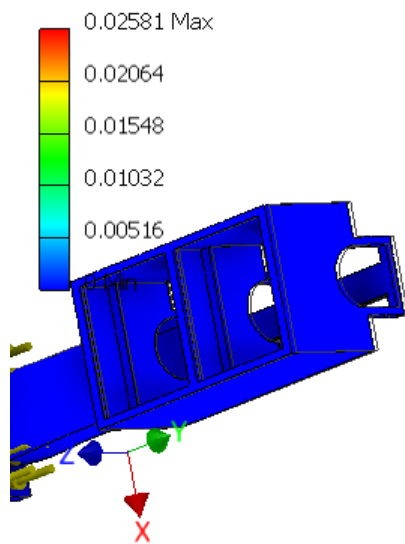
3rd Principal Stress

Type: 3rd Principal Stress
Unit: MPa
2026-08-14, 오후 1:05:41



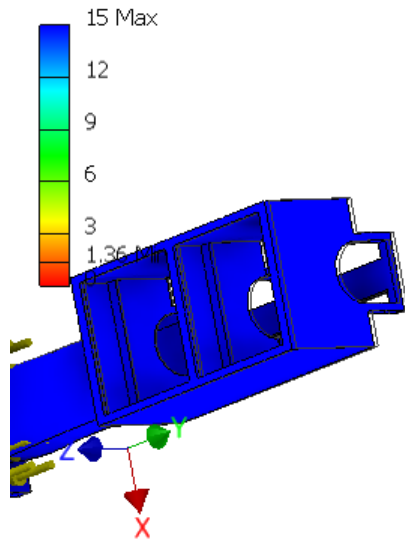
Displacement

Type: Displacement
Unit: mm
2026-08-14, 오후 1:05:41



Safety Factor

Type: Safety Factor
Unit: ul
2026-08-14, 오후 1:05:41



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